ADVANCED SPLUNK ADMIN & DEVELOPER

Beginner-Friendly Hands-on Course with Splunk Enterprise

90 hours (4.5 months) • Splunk Enterprise Trial • Administration • SPL • Dashboards • Data Onboarding • Automation

TOTAL COURSE DURATION: 90 HOURS

Practical / Hands-On

Recommended delivery: 30 sessions × 3 hours (or equivalent instructor-led schedule). Module hours include instructor explanation, demonstrations, guided labs, troubleshooting and short assessments.

Course Overview

A practical beginner-to-intermediate course designed around a self-managed Splunk Enterprise lab. Participants learn the platform from first installation through data onboarding, SPL searching, knowledge objects, dashboards, alerts, administration, troubleshooting and developer-oriented automation.

The current Splunk Enterprise installation provides a 60-day Enterprise Trial by default, with access to Enterprise features during the trial and a 500 MB/day indexing limit. After the trial, the installation can convert to Splunk Free, which has significant feature restrictions. Therefore, the labs are designed to be completed within the 60-day trial window.

Official Splunk documentation confirms these trial conditions.

Target Audience

- Cybersecurity beginners and fresh graduates
- SOC analysts moving toward Splunk administration
- System/network administrators entering SIEM
- Security engineers and junior Splunk engineers
- IT professionals who want practical SPL and dashboard skills

Prerequisites

- Basic computer and networking knowledge
- Basic understanding of Windows/Linux logs is helpful but not mandatory
- No previous Splunk experience required
- A laptop/VM capable of running Splunk Enterprise
- Splunk.com account for the Enterprise Trial

Recommended Lab Setup

- Splunk Enterprise latest supported trial build
- Windows or Linux host/VM
- Chrome/Edge/Firefox browser
- Optional: Windows and Linux test VMs for log generation
- Sample/tutorial datasets for early exercises
- Optional: Splunk Universal Forwarder for data collection labs
- Optional: Sysmon for richer Windows security telemetry

60-Day Learning Roadmap

Phase	Suggested Days	Focus
Foundation	1–10	Install, UI, architecture, data model, first searches
SPL & Investigation	11–25	Search language, fields, statistics, lookups, time
Developer Skills	26–35	Knowledge objects, field extraction, dashboards, macros
Administration	36–48	Indexes, inputs, users, apps, configuration, monitoring
Advanced Operations	49–56	Forwarders, clustering concepts, troubleshooting, APIs
Capstone	57–60	End-to-end SIEM implementation and presentation

Module Duration Summary

Module 01 — Splunk Fundamentals & Platform Orientation (3 Hours)

Module Focus: Understand what Splunk does, its core components, terminology and the data-to-insight workflow.

Theory Concepts

- What is Splunk and machine data
- Events, fields, timestamps and metadata
- Search, index and visualization concepts
- Splunk Enterprise vs Splunk Cloud overview
- Basic Splunk architecture
- Search head, indexer and forwarder roles
- Indexes and buckets at a conceptual level
- Splunk Web navigation and apps

Hands-on Lab

Lab Objective: Get comfortable with the Splunk interface and understand the journey from raw log to search result.

Lab Activities

1. Log in to Splunk Web and inspect the home/search interfaces.
2. Explore Settings, Apps, Search & Reporting and Monitoring Console.
3. Identify the roles of search head, indexer and forwarder using a simple architecture diagram.
4. Load the tutorial/sample dataset.

5. Run a first search and inspect events, fields and timestamps.

Splunk / Lab Tools

- Splunk Enterprise Trial
- Browser
- Splunk tutorial data

Expected Outcome

Participant can explain Splunk's purpose and identify the major platform components.

Module 02 — Installation, Licensing & Initial Configuration (4 Hours)

Module Focus: Install Splunk Enterprise correctly and prepare a stable learning environment.

Theory Concepts

- Enterprise Trial licensing
- Installation options on Windows/Linux
- Splunk Web and management services
- Initial administrator setup
- Ports and service basics
- License volume and 500 MB/day trial limit
- Health checks and restart concepts
- Security basics for the lab

Hands-on Lab

Lab Objective: Install and initialize a standalone Splunk Enterprise instance.

Lab Activities

6. Create/login to a Splunk.com account and obtain the Enterprise Trial.
7. Install Splunk Enterprise on Windows or Linux.
8. Launch Splunk Web and complete initial configuration.
9. Check license status and record the trial start date.
10. Verify service status and basic network connectivity.
11. Create a lab checklist for daily license-volume monitoring.

Splunk / Lab Tools

- Splunk Enterprise Trial
- Windows/Linux
- Browser

Expected Outcome

Participant can install Splunk Enterprise and verify that the instance is ready for labs.

Module 03 — Data Ingestion & Inputs (4 Hours)

Module Focus: Learn how data enters Splunk and how to choose an appropriate input method.

Theory Concepts

- Data ingestion pipeline
- File and directory monitoring

- Network inputs concept
- Universal Forwarder concept
- Event breaking and timestamps
- Source, sourcetype and host
- Index selection
- Input configuration fundamentals

Hands-on Lab

Lab Objective: Bring multiple types of sample data into Splunk and classify them correctly.

Lab Activities

12. Upload a sample log file through Splunk Web.
13. Configure a monitored file/directory input in the lab.
14. Review host, source and sourcetype values.
15. Send a second dataset into a separate index.
16. Compare the resulting metadata.
17. Troubleshoot one intentionally misconfigured input.

Splunk / Lab Tools

- Splunk Enterprise
- Sample logs
- Optional Universal Forwarder

Expected Outcome

Participant can onboard basic data and explain source, sourcetype, host and index.

Module 04 — Search Fundamentals & SPL Basics (5 Hours)

Module Focus: Build confidence with Splunk Search Processing Language through small, understandable searches.

Theory Concepts

- SPL pipeline concept
- Search terms and keywords
- Time ranges
- Fields and field-value searches
- AND/OR/NOT
- Wildcards
- Comments and formatting
- Search Job Inspector basics

Hands-on Lab

Lab Objective: Write and modify beginner SPL searches against the training data.

Lab Activities

18. Search for all events in a selected index.
19. Filter by host, source and sourcetype.
20. Use time ranges to narrow results.
21. Search using field-value conditions.
22. Combine conditions with Boolean operators.
23. Inspect search performance and results.

Splunk / Lab Tools

- Splunk Search & Reporting
- Splunk Search Reference

Expected Outcome

Participant can write basic SPL and confidently narrow a dataset to relevant events.

Module 05 — Fields, Events & Data Enrichment (4 Hours)

Module Focus: Learn how Splunk identifies useful fields and how analysts can improve event usability.

Theory Concepts

- Automatic field extraction
- Indexed-time vs search-time concepts
- Field discovery
- Field aliases and calculated fields
- Tags and event types
- Lookup enrichment concepts
- Data normalization basics

Hands-on Lab

Lab Objective: Identify and improve fields in a sample security dataset.

Lab Activities

24. Inspect automatically extracted fields.
25. Create a simple calculated field or alias in the lab.
26. Create tags for selected values.
27. Create an event type for a recurring event pattern.
28. Apply a lookup to enrich selected events.
29. Validate the enriched results with SPL.

Splunk / Lab Tools

- Splunk Knowledge Objects
- CSV lookup file
- Search & Reporting

Expected Outcome

Participant can explain fields and create basic knowledge objects for analysis.

Module 06 — SPL for Security Investigation (5 Hours)

Module Focus: Use statistics and transformation commands to turn raw events into useful security findings.

Theory Concepts

- stats and charting concepts
- count, dc, values and list
- timechart
- top and rare
- sort, dedup and rename
- eval and conditional logic

- where and search differences
- transaction concept and alternatives

Hands-on Lab

Lab Objective: Investigate authentication and network-style events using SPL.

Lab Activities

30. Find the most active users.
31. Count events by source IP.
32. Identify users with multiple failed logins.
33. Trend events over time with timechart.
34. Use eval to classify events.
35. Build a search that highlights unusual activity patterns.
36. Save the investigation search for reuse.

Splunk / Lab Tools

- Splunk SPL
- Training security logs

Expected Outcome

Participant can transform raw logs into investigation-ready statistics.

Module 07 — Advanced SPL & Search Optimization (5 Hours)

Module Focus: Develop cleaner, faster and more maintainable searches.

Theory Concepts

- Subsearch concepts
- Joins and their limitations
- Append and appendcols concepts
- Lookup and lookup enrichment
- mvexpand and multivalue fields
- rex for extraction
- Regular expressions
- Search optimization principles
- Job Inspector and performance awareness

Hands-on Lab

Lab Objective: Build several investigation searches and improve their efficiency.

Lab Activities

37. Extract a field using rex.
38. Enrich events using a lookup.
39. Compare a simple search with a less efficient approach.
40. Use Job Inspector to inspect search behavior.
41. Rewrite a search to reduce unnecessary processing.
42. Document why the optimized version is preferable.

Splunk / Lab Tools

- SPL
- rex/regex

- Job Inspector
- Lookup files

Expected Outcome

Participant can build more advanced SPL and recognize common performance problems.

Module 08 — Time, Buckets & Data Lifecycle (4 Hours)

Module Focus: Understand how Splunk stores events and why time, retention and bucket concepts matter to administrators.

Theory Concepts

- Event time vs index time
- Hot, warm, cold and frozen bucket concepts
- Bucket lifecycle
- Retention concepts
- Index sizing fundamentals
- Time modifiers in SPL
- Earliest/latest and relative time
- Data volume awareness

Hands-on Lab

Lab Objective: Explore time-based searching and simulate an administrator's data-retention planning exercise.

Lab Activities

43. Compare searches using relative time modifiers.
44. Inspect event timestamps and metadata.
45. Review index configuration concepts.
46. Calculate an example daily ingestion and retention requirement.
47. Explain how bucket lifecycle supports retention.
48. Create a simple capacity-planning worksheet.

Splunk / Lab Tools

- Splunk Web
- SPL
- Index configuration

Expected Outcome

Participant can explain Splunk's bucket lifecycle and perform basic retention/capacity reasoning.

Module 09 — Indexes & Index Administration (4 Hours)

Module Focus: Learn how administrators organize data and control storage and search behavior.

Theory Concepts

- Index purpose
- Default and custom indexes
- Index naming conventions
- Home, cold and thawed paths concept
- Retention settings

- Data model planning
- Permissions and index access
- Index troubleshooting basics

Hands-on Lab

Lab Objective: Create and administer a dedicated training index.

Lab Activities

49. Create a custom index for security training data.
50. Configure appropriate event/data settings.
51. Send selected data to the new index.
52. Verify searchability and permissions.
53. Review index health and storage information.
54. Document an index design standard for future projects.

Splunk / Lab Tools

- Splunk Enterprise
- Settings > Indexes
- CLI (optional)

Expected Outcome

Participant can create a basic index and explain its role in Splunk data management.

Module 10 — Users, Roles & Authentication (3 Hours)

Module Focus: Understand how access control is implemented in Splunk Enterprise.

Theory Concepts

- Users and roles
- Capabilities and permissions
- Role inheritance
- Index-based access
- Authentication concepts
- Local authentication
- LDAP/SSO concepts
- Least privilege

Hands-on Lab

Lab Objective: Create a small role model for a training SOC.

Lab Activities

55. Create a viewer/analyst-style role for the lab.
56. Create a test user and assign the role.
57. Restrict access to a training index.
58. Test allowed and denied actions.
59. Review capability settings.
60. Document a least-privilege access model.

Splunk / Lab Tools

- Splunk Enterprise user/role management

Expected Outcome

Participant can explain Splunk authorization and implement basic least-privilege access.

Module 11 — Configuration Files & Splunk Administration (4 Hours)

Module Focus: Introduce the configuration layer used by Splunk administrators.

Theory Concepts

- Splunk configuration architecture
- conf files and precedence concepts
- Inputs configuration
- Indexes configuration
- Props and transforms overview
- Deployment considerations
- CLI vs Web configuration
- Backup and change-management practices

Hands-on Lab

Lab Objective: Inspect and safely modify selected configuration settings in the lab.

Lab Activities

61. Locate the relevant Splunk configuration directories.
62. Inspect selected .conf files without changing defaults blindly.
63. Identify the purpose of props.conf, transforms.conf and inputs.conf.
64. Make a controlled lab configuration change.
65. Restart/reload as appropriate.
66. Verify the resulting behavior and document the change.

Splunk / Lab Tools

- Splunk Enterprise CLI
- splunk command
- Text editor/VS Code

Expected Outcome

Participant can identify major Splunk configuration files and understand safe configuration management.

Module 12 — Universal Forwarder & Distributed Data Collection (5 Hours)

Module Focus: Understand how Splunk Universal Forwarder supports scalable log collection.

Theory Concepts

- Forwarder role
- Universal Forwarder vs Heavy Forwarder
- Receiving and forwarding concepts
- Deployment Server concept
- inputs.conf basics
- outputs.conf basics
- Forwarding security concepts

- Troubleshooting data flow

Hands-on Lab

Lab Objective: Connect a Universal Forwarder to the training Splunk instance.

Lab Activities

67. Install the Universal Forwarder on a lab endpoint.
68. Configure the target Splunk receiver.
69. Monitor a selected log file.
70. Verify the data arrives with correct metadata.
71. Stop the input and troubleshoot missing data.
72. Document the data path from source to index.

Splunk / Lab Tools

- Splunk Universal Forwarder
- Splunk Enterprise
- Windows/Linux lab VM

Expected Outcome

Participant can explain and implement a basic forwarder-to-indexer data path.

Module 13 — Knowledge Objects & Field Extraction (4 Hours)

Module Focus: Build reusable objects that make searches consistent for analysts.

Theory Concepts

- Knowledge object lifecycle
- Field extractions
- Props and transforms relationship
- Calculated fields
- Aliases
- Tags and event types
- Lookup definitions
- Object ownership and permissions

Hands-on Lab

Lab Objective: Create a small reusable knowledge layer for security logs.

Lab Activities

73. Create a field extraction for a lab log format.
74. Create an alias or calculated field.
75. Build an event type for a security event.
76. Add a tag to selected values.
77. Create a lookup and apply it to a search.
78. Review object permissions.

Splunk / Lab Tools

- Splunk Web
- props.conf
- transforms.conf
- Lookup CSV

Expected Outcome

Participant can create and manage reusable Splunk knowledge objects.

Module 14 — Reports, Alerts & Scheduled Searches (3 Hours)

Module Focus: Turn searches into repeatable monitoring workflows.

Theory Concepts

- Reports vs alerts
- Scheduled searches
- Alert conditions
- Trigger actions
- Throttle concepts
- Email notification concept
- Risk of noisy alerts
- Alert testing and troubleshooting

Hands-on Lab

Lab Objective: Create useful security alerts without generating unnecessary noise.

Lab Activities

79. Create a report for repeated authentication failures.
80. Convert a search into a scheduled alert.
81. Define a sensible threshold and time window.
82. Configure a lab-safe notification/action.
83. Test the alert using generated training events.
84. Tune the condition to reduce false positives.

Splunk / Lab Tools

- Splunk Enterprise Alerts
- Search & Reporting

Expected Outcome

Participant can create, test and tune a basic alert.

Module 15 — Dashboards & Visual Analytics (4 Hours)

Module Focus: Create analyst-friendly dashboards that summarize security and operational data.

Theory Concepts

- Dashboard design principles
- Panels and searches
- Single value, table, chart and timechart
- Tokens and filters
- Drilldowns
- Simple XML concepts
- Dashboard usability
- Performance considerations

Hands-on Lab

Lab Objective: Build a security monitoring dashboard.

Lab Activities

85. Create panels for event volume, top users, source IPs and authentication failures.
86. Add a time-series panel.
87. Add a time-range input/token.
88. Configure a drilldown from a summary panel to detailed events.
89. Test the dashboard with different time windows.
90. Document the purpose of each panel.

Splunk / Lab Tools

- Splunk Dashboard Studio or Simple XML
- Search & Reporting

Expected Outcome

Participant can build a functional beginner-friendly monitoring dashboard.

Module 16 — Lookups, Macros, Event Types & Tags (3 Hours)

Module Focus: Develop reusable building blocks that simplify searches and improve consistency.

Theory Concepts

- CSV and KV Store lookup concepts
- Automatic lookups
- Search macros
- Event types
- Tags
- Object naming standards
- Sharing and permissions

Hands-on Lab

Lab Objective: Build reusable enrichment and search components.

Lab Activities

91. Create a small asset or user lookup.
92. Use the lookup in SPL.
93. Create a search macro for a repeated filter.
94. Create an event type for a common security pattern.
95. Apply tags to important values.
96. Use the objects in a dashboard/report.

Splunk / Lab Tools

- Splunk Web
- CSV lookup
- SPL

Expected Outcome

Participant can create reusable search components and enrich investigations.

Module 17 — Monitoring Console & Splunk Health (3 Hours)

Module Focus: Learn how an administrator monitors Splunk itself.

Theory Concepts

- Monitoring Console purpose
- System health
- Indexing performance
- Search performance
- Resource utilization
- License monitoring
- Internal indexes
- Troubleshooting workflow

Hands-on Lab

Lab Objective: Use Monitoring Console and internal logs to identify a simulated platform problem.

Lab Activities

97. Open Monitoring Console.
98. Review indexing and search health.
99. Check license volume.
100. Inspect selected internal logs.
101. Simulate a common lab issue such as a stopped input.
102. Identify symptoms, evidence and likely cause.
103. Document corrective action.

Splunk / Lab Tools

- Monitoring Console
- _internal
- _introspection

Expected Outcome

Participant can perform a basic Splunk health check and start a structured troubleshooting investigation.

Module 18 — Splunk REST API & Developer Fundamentals (4 Hours)

Module Focus: Introduce programmatic interaction with Splunk for developers and automation engineers.

Theory Concepts

- REST API concepts
- Authentication/session concepts
- Endpoints and resources
- Search jobs
- Exporting results
- JSON response structure
- API security basics
- Automation use cases

Hands-on Lab

Lab Objective: Use a simple script to interact with the local Splunk instance.

Lab Activities

104. Authenticate to the local API using a lab account.
105. Create or submit a safe search job.
106. Poll for completion.
107. Retrieve structured results.
108. Parse JSON in Python.
109. Handle errors and timeouts safely.

Splunk / Lab Tools

- Splunk REST API
- Python
- Requests
- JSON

Expected Outcome

Participant can explain the Splunk API workflow and build a basic read-oriented automation script.

Module 19 — SPL for SOC Use Cases (5 Hours)

Module Focus: Apply SPL to common security monitoring scenarios.

Theory Concepts

- Authentication monitoring
- Brute-force detection logic
- Suspicious process/logon concepts
- Unusual source IP analysis
- Data-exfiltration indicators at a conceptual level
- Windows event analysis
- Threat hunting workflow
- MITRE ATT&CK mapping

Hands-on Lab

Lab Objective: Create a mini library of SOC searches.

Lab Activities

110. Build a failed-login detection search.
111. Identify successful login after repeated failures.
112. Find unusual login sources in the training dataset.
113. Build a simple suspicious-process search using available logs.
114. Map selected detections to MITRE ATT&CK techniques.
115. Create a dashboard panel for the use case.

Splunk / Lab Tools

- Splunk SPL
- Windows/Linux sample logs
- MITRE ATT&CK

Expected Outcome

Participant can translate common SOC questions into practical SPL searches.

Module 20 — Troubleshooting & Operational Best Practices (4 Hours)

Module Focus: Develop a repeatable approach for diagnosing Splunk issues.

Theory Concepts

- Data missing from search
- Wrong sourcetype/index
- Timestamp problems
- Parsing and field extraction issues
- Forwarder connectivity
- License warnings
- Search performance problems
- Service restart considerations
- Evidence-based troubleshooting

Hands-on Lab

Lab Objective: Troubleshoot a set of instructor-created Splunk problems.

Lab Activities

116. Diagnose a missing-data scenario.
117. Find a metadata mismatch.
118. Identify a timestamp/parsing problem.
119. Troubleshoot a forwarder/input issue.
120. Investigate a slow search using Job Inspector.
121. Write a root-cause and remediation note for each issue.

Splunk / Lab Tools

- Splunk Web
- CLI
- Monitoring Console
- _internal logs
- Job Inspector

Expected Outcome

Participant can follow a structured troubleshooting methodology instead of trial-and-error changes.

Module 21 — Architecture, Clustering & Scale Concepts (3 Hours)

Module Focus: Prepare learners for real enterprise Splunk environments without requiring a production cluster.

Theory Concepts

- Single-instance architecture
- Distributed search
- Search head clustering concept
- Indexer clustering concept
- Cluster manager concept
- Deployment server concept
- Heavy forwarder concept
- High availability and scalability

- Capacity planning

Hands-on Lab

Lab Objective: Design an enterprise-style Splunk architecture on paper and relate it to the lab instance.

Lab Activities

122. Draw a standalone architecture.
123. Extend it into a distributed architecture.
124. Assign search, indexing and collection roles.
125. Explain data flow between components.
126. Identify high-availability points.
127. Discuss what cannot be reproduced on a single trial VM.

Splunk / Lab Tools

- Architecture diagrams
- Splunk Enterprise trial lab

Expected Outcome

Participant can explain enterprise Splunk architecture and the purpose of major distributed components.

Module 22 — Final Capstone — Build a Mini SOC Splunk Environment (7 Hours)

Module Focus: Combine administration, SPL and developer skills into one portfolio project.

Theory Concepts

- Project scope and requirements
- Data onboarding
- Index and sourcetype design
- SPL investigations
- Knowledge objects
- Alerts
- Dashboard design
- Access control
- Troubleshooting
- Documentation

Hands-on Lab

Lab Objective: Build a small security-monitoring environment from raw data to dashboard and alert.

Lab Activities

128. Install and validate Splunk Enterprise.
129. Create a dedicated security index.
130. Onboard at least two datasets.
131. Create useful field extractions/lookups.
132. Develop 5–8 SOC searches.
133. Create 2–3 alerts.
134. Build a security dashboard.
135. Create analyst/viewer roles.
136. Troubleshoot one intentionally injected problem.

137. Prepare a final architecture, SPL and administration report.

Splunk / Lab Tools

- Splunk Enterprise Trial
- Optional Universal Forwarder
- Python/REST API
- Sample security logs

Expected Outcome

Participant finishes with a complete beginner-friendly Splunk project demonstrating both administration and developer skills.

Suggested Final Deliverables

- Splunk lab architecture diagram
- Index and data-onboarding document
- SPL search library
- Field-extraction / lookup documentation
- SOC dashboard
- Alert definitions and tuning notes
- User/role access matrix
- Troubleshooting runbook
- REST API / Python mini-script
- Final capstone presentation

60-Day Trial Management Tips

- Start the trial only when the learner is ready to begin the practical lab period.
- Keep ingestion below the trial's documented daily volume limit.
- Use compact sample datasets for exercises instead of continuously generating large logs.
- Reserve the final week for the capstone and troubleshooting practice.
- Track the trial start date and license status from the first session.
- If the trial expires, Splunk Free may remain available for standalone learning, but it has feature restrictions and should not be treated as equivalent to the Enterprise Trial.

Official Reference Notes

The course design is aligned with current official Splunk documentation for installation, trial licensing, data onboarding, SPL, administration and developer resources.

- Splunk Enterprise Trial: 60-day Enterprise Trial with up to 500 MB/day indexing during the trial.
- Splunk Enterprise can be installed on supported Windows and Linux environments.
- Splunk's official Search Tutorial uses tutorial data and teaches data ingestion, SPL and dashboards.
- Splunk documentation covers administration, data onboarding, knowledge objects, REST API and architecture.

Official references used for course planning: Splunk Enterprise documentation and Splunk Search Tutorial.

Course Positioning

SPLUNK ADMIN & DEVELOPER

From Splunk Beginner → SPL Practitioner → Administrator → Developer

Module	Module Title	Hours
01	Splunk Fundamentals & Platform Orientation	3
02	Installation, Licensing & Initial Configuration	4
03	Data Ingestion & Inputs	4
04	Search Fundamentals & SPL Basics	5
05	Fields, Events & Data Enrichment	4
06	SPL for Security Investigation	5
07	Advanced SPL & Search Optimization	5
08	Time, Buckets & Data Lifecycle	4
09	Indexes & Index Administration	4
10	Users, Roles & Authentication	3
11	Configuration Files & Splunk Administration	4
12	Universal Forwarder & Distributed Data Collection	5
13	Knowledge Objects & Field Extraction	4
14	Reports, Alerts & Scheduled Searches	3
15	Dashboards & Visual Analytics	4
16	Lookups, Macros, Event Types & Tags	3
17	Monitoring Console & Splunk Health	3
18	Splunk REST API & Developer Fundamentals	4
19	SPL for SOC Use Cases	5
20	Troubleshooting & Operational Best Practices	4
21	Architecture, Clustering & Scale Concepts	3
22	Final Capstone — Build a Mini SOC Splunk Environment	7
	TOTAL COURSE DURATION	90

Planning note: The 90 hours are distributed according to the practical depth of the existing 22-module course. The existing 60-day Enterprise Trial roadmap remains a lab/licensing window, not a 60-hour teaching requirement.